

PHYS 4730 – Computational Physics: Statistics and Data Science

Lab01 – Course Introduction

2026/08/25

UU Computational Physics Sequence

- PHYS 2235 – Introduction to Python and Linux
- PHYS 3010 lab – More Python, including general tools like NumPy and SciPy. Symbolic manipulation e.g. Maple, SymPy
- **PHYS 4730 – Advanced topics with a data science focus.**
- PHYS 5730 – Advanced topics with a focus on algorithms and simulations.

Physics 4730, Computational Physics: Statistics and Data Science (Fall 2026)

Instructor	John Belz	LSSB W2205 office hours: by appointment jbr0jb@gmail.com
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Description	This advanced course is focused on development of a skillset suitable for the analysis and statistical interpretation of scientific data. Students may write programs in both general-purpose (Python, C++ ...) and symbol manipulation languages (SymPy, Mathematica ...). Specific topics include data mining, basic probability distributions (binomial, Poisson, Gaussian), error propagation, parameter estimation by χ^2 minimization and maximum likelihood, comparison of distributions, Bayesian inference, machine learning and neural networks. Current "hot topics" may be presented at the survey level. Exercises will be chosen to illustrate methods applicable to active research topics. This is a required course for undergraduate students working towards the Computational Physics Emphasis.	
Prerequisites	PHYS 3010 AND (MATH 2250 OR (MATH 2270 AND MATH 2280))	
Lectures/Labs	TTh 08:35 – 10:30 LSSB W3262	
Web Page	https://jwbelz.github.io/phys4730/	
Text	There is no textbook for the course; All supplementary information can be obtained online.	
Grading	Grading for the course will break down as follows:	
	In-Class Exercises	25%
	Homework	50%
	Final Project	25%
Drop/Add	Last day to add, August 28. Last day to drop, September 4. Last day to withdraw, November 11.	

Lecture and Lab Schedule – Physics 4730, Fall 2026

Week 01	08/25	Course Introduction
	08/27	Python and Linux Review
Week 02	09/01	Linear Algebra, Pseudoinverse
	09/03	Random Variables
Week 03	09/08	Basic probability distributions. Binomial, Poisson, Gaussian
	09/10	Error Propagation and Covariance
Week 04	09/15	Parameter Estimation: Method of Moments
	09/17	Parameter Estimation: χ^2 Minimization
Week 05	09/22	Parameter Estimation: Maximum Likelihood I
	09/24	Maximum Likelihood II
Week 06	09/29	χ^2 Minimization II: Goodness-of-Fit
	10/01	Fisher Information
Week 07	10/06	Confidence Intervals
	10/08	Empirical Maximum Likelihood I
Week 08	10/13	NO CLASS, FALL BREAK
	10/15	NO CLASS, FALL BREAK
Week 09	10/20	Empirical Maximum Likelihood II
	10/22	Comparing Distributions: KS Test
Week 10	10/27	Comparing 2D RV Distributions
	10/29	Comparing Distributions with Unknown Parameters
Week 11	11/03	Final Project Assignment
	11/05	Bayes' Theorem
Week 12	11/10	Introduction to Machine Learning
	11/12	Decision Trees
Week 13	11/17	Dealing with Real Data, Nearest Neighbor Algorithm
	11/19	Diagnostics and Evaluation of Supervised Learning
Week 14	11/24	Neural Network Basics
	11/26	NO CLASS, THANKSGIVING HOLIDAY
Week 15	12/01	More Neural Networks
	12/03	Neural Networks and Data Analysis
Week 16	12/08	Final project help
	12/10	Final project help

Today

- Log-in to physics cluster (group activity)
- Set **.forward** to receive grade reports
- Edit a file
- Confirm **submit** utility working

Login Workflow

01 Open Windows App

Launch the remote desktop app.

02 Add a PC

Use the + button to add a new device.

03 Enter the server

Choose canis, lyra, or ursa.

04 ~~Authenticate~~

~~Use your University username and CIS password.~~

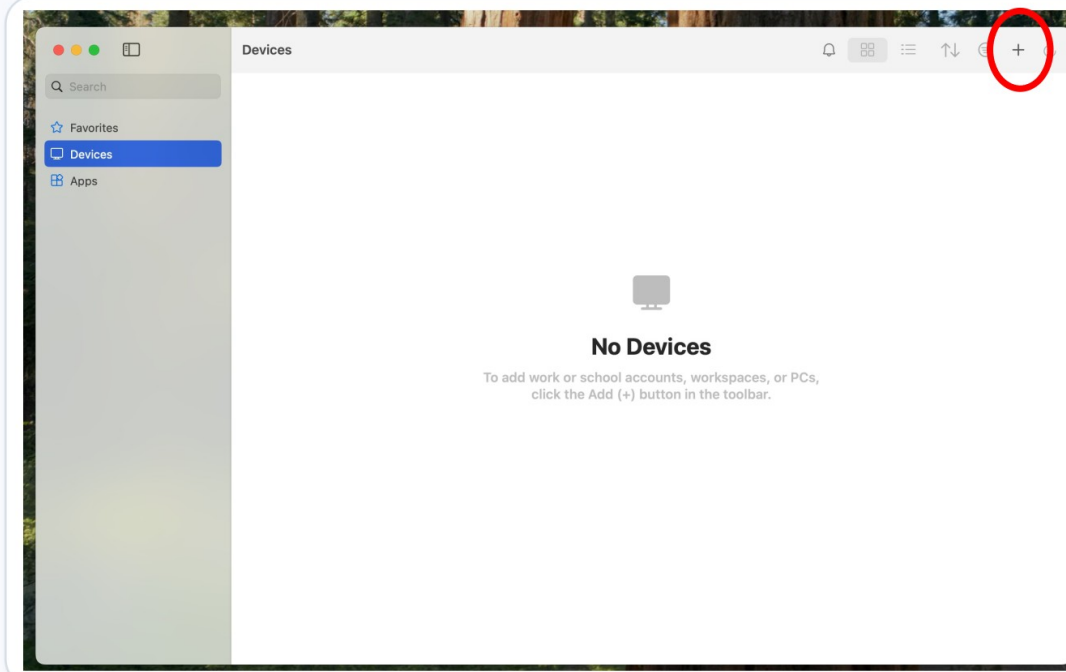
05 Open Terminal

Start working from the Linux desktop.

Server names: canis.physics.utah.edu • lyra.physics.utah.edu • ursa.physics.utah.edu

Use **physics.utah.edu** credentials!

Open The Remote Desktop App



1 Find the Devices view

You should see the option to add a new PC or workspace.

2 Click +

The next screen asks for the PC name.

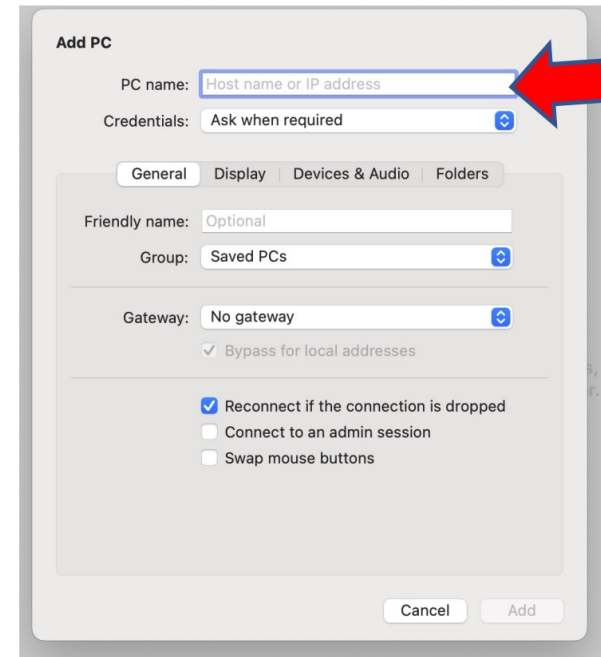
Enter the Server Name

Choose one of the approved Physics servers:

canis.physics.utah.edu

lyra.physics.utah.edu

ursa.physics.utah.edu



The screenshot shows the 'Add PC' dialog box with the following fields and options:

- PC name:** Host name or IP address (indicated by a red arrow)
- Credentials:** Ask when required
- Tabs:** General, Display, Devices & Audio, Folders
- Friendly name:** Optional
- Group:** Saved PCs
- Gateway:** No gateway
- ☒ Bypass for local addresses
- ☒ Reconnect if the connection is dropped
- ☐ Connect to an admin session
- ☐ Swap mouse buttons
- Buttons:** Cancel, Add

Save the Connection

Add PC

PC name:

Credentials:

General | Display | Devices & Audio | Folders

Friendly name:

Group:

Gateway:

☒ Bypass for local addresses

☒ Reconnect if the connection is dropped

☐ Connect to an admin session

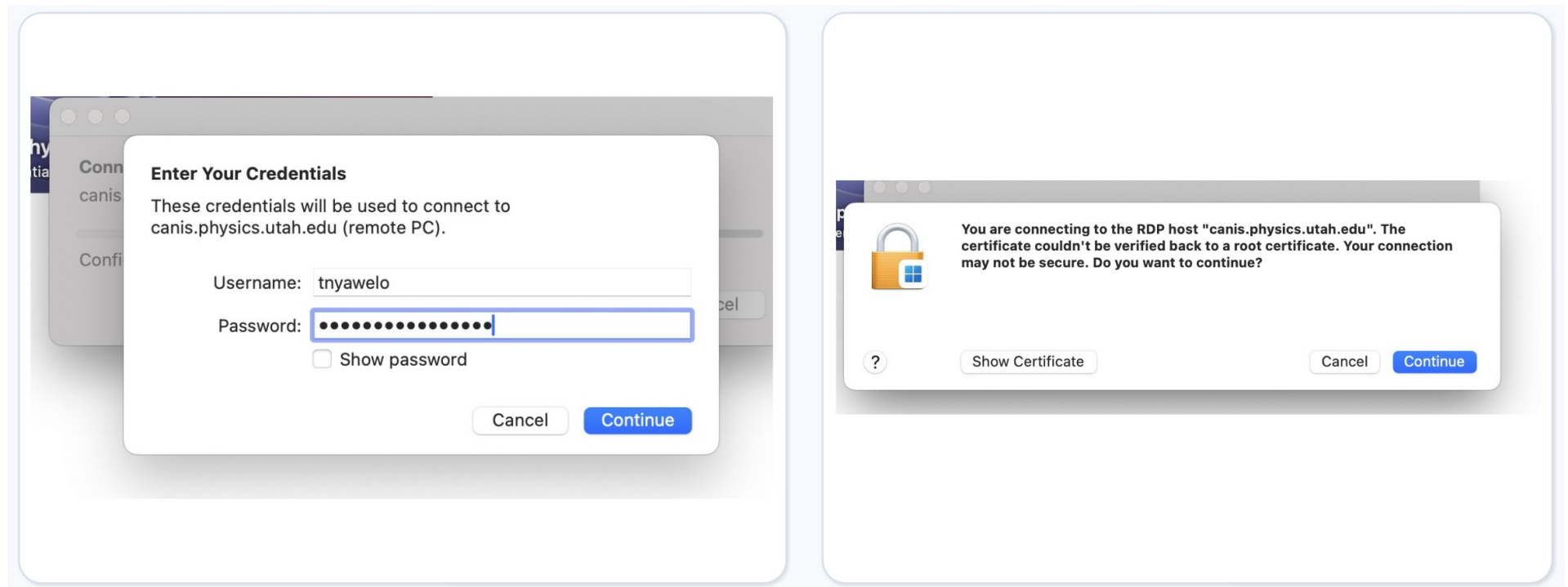
☐ Swap mouse buttons



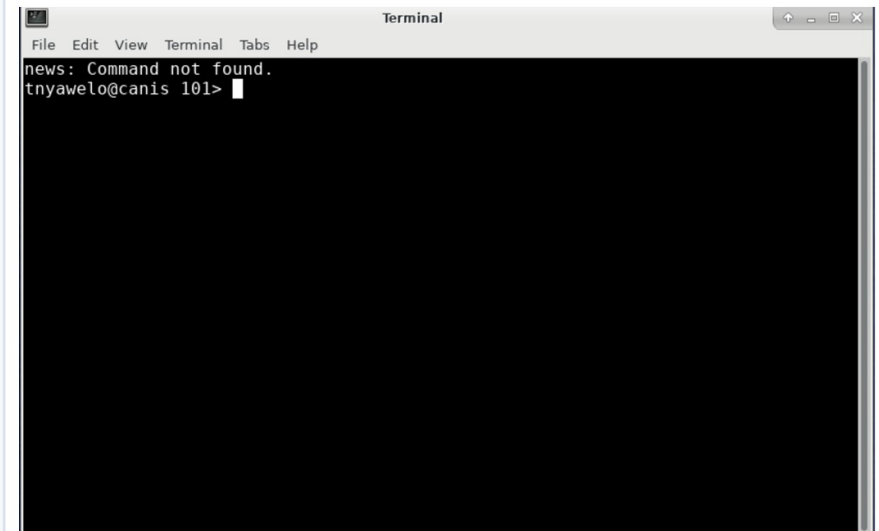
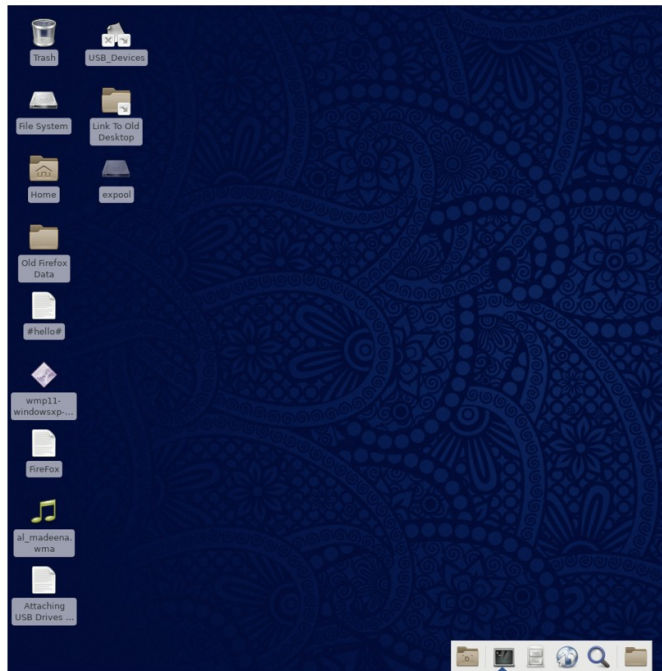
Hide Sideb

After adding the PC, the saved device appears in your workspace.

Authenticate & Continue



Open a terminal



Open a Terminal from the Linux desktop. You should see a command prompt.

Today's Assignment

Mandatory Assignment, due 25 August 2026 in class

Problem 1

The assignments, exams, and projects for this course will be submitted electronically using the **submit** utility. The purpose of this exercise is to practice using this utility.

Some general instructions for the **submit** system are:

- The general form of a command is **submit p4730 [assignment] [file1] [file2] [file3]...**
- If you type only **submit p4730** you will get a list of pending assignments. Try it!
- If you type **submit p4730 [assignment]** you will get a listing of files you have submitted for this assignment. Of course, if you haven't yet submitted anything the list will be empty.
- You may submit several files at once, or one at a time. You can resubmit files up until the due date. Note that if you resubmit a file, the original submitted version of that file will be lost.
- A common pitfall of this system is submitting files with names that do not **exactly** match what the assignment requires. In Linux, remember that file names are case-sensitive.

Now, please perform the following:

1. Create a file called **exactly**

Lab01_Problem01.txt

2. In this file, provide brief responses to the following:

(a) What was the highlight of your summer break?

(b) Please describe your previous computing experience. List any prior programming courses that you have taken. List any programming languages and operating systems with which you are familiar.

3. Submit your file for grading with the command:

```
submit p4730 lab01 Lab01_Problem01.txt
```

4. Check your submission with the command:

```
submit p4730 lab01
```

You should get a notice that your file has been received.

5. Await your grade report via email (may take a few days...)